

Midterm Exam for “The Mathematical Foundations of Political Methodology”

You may use a simple (non-graphing) calculator and refer to your Pemberton & Rau textbook and your class notes. Please do not use the internet during the exam. Show your work and *be tidy*!

1. Given the following hold, what (if anything) do we know about the relationship between sets A and C ?

(a) $A \cap B = A$ and $B \cap C \neq \emptyset$

(b) $A \cup B = A$ and $B \cap C \neq \emptyset$

(c) $C \cup D = B$ and $B \cap C = A$

2. Find the limit of each of the following sequences as $n \rightarrow \infty$, if the limit exists.

(a)

$$\frac{4n^3 - 1}{(n + 1)^3}$$

(b)

$$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{2}} + n^{\frac{2}{5}}}$$

3. Use the (δ, ϵ) definition of a limit to prove that the function

$$f(x) = \frac{1}{2}x - 10$$

is everywhere continuous.

4. Let

$$f(x) = \frac{x^2}{(x + 4)}$$

- (a) Find the critical points of f and characterize them as maxima, minima, or inflection points.
- (b) Find the regions on which f is increasing and decreasing.
- (c) Find any asymptotes that f may have.
- (d) Use your answers above to roughly plot out the shape of f . It doesn't have to be perfect!

5. Perform the following matrix operations:

(a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Find the rank of

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 2 & 4 \\ -2 & 0 & -6 \\ 1 & 9 & 21 \end{bmatrix}$$

6. Find and classify all of the critical points of the following functions. Hint: Some of these functions may be concave or convex, in which case you only need to find the critical points to classify them.

(a)

$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

(b)

$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

(c)

$$f(x, y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$